

# Initial Reading Notes on SABR Models for Synthetic FX Rates

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## 1. Objective of the paper

The paper studies how to reconstruct the smile of an illiquid cross FX rate from two more liquid FX pairs quoted against a common base currency. The main result is that, up to the same asymptotic order as the usual SABR formula, the cross can be represented by an effective lognormal SABR model.

## 2. Global understanding

- Main text: formulate the cross FX problem and state the effective SABR result in practical form.
- Appendix A: derive the effective forward equation for a general multifactor stochastic-volatility model.
- Appendix B: specialize the general formulas to the cross FX case.
- Appendix C: recall standard SABR implied-volatility formulas.
- Main mathematical idea: reduce a multi-factor model to a one-dimensional marginal description, then identify that description with SABR.

## 3. Part I focused on

I chose to focus especially on **Appendix A**.

- It is the mathematical core of the paper.
- It explains *why* an effective SABR model appears, not only *what* the final parameters are.
- Once Appendix A is understood, Appendix B becomes much easier to read as a specialization to cross FX.

## 4. What I understood well

- The cross FX rate is written as a ratio of two forwards:

$$\tilde{X}_{12} = \frac{\tilde{X}_{10}}{\tilde{X}_{20}}.$$

- Under the appropriate forward measure, the drift disappears and

$$\frac{d\tilde{X}_{12}}{\tilde{X}_{12}} = \varepsilon \left( \tilde{A}_1 dW_1 - \tilde{A}_2 dW_2 \right).$$

- The cross is therefore driven by a *difference* of liquid FX drivers.
- The instantaneous cross variance is controlled by

$$\Delta^2 = A_1^2 - 2cA_1A_2 + A_2^2,$$

which already suggests an effective volatility factor.

- Appendix A builds a closure relation between the marginal density  $Q^{(0)}$  and the weighted density  $Q^{(2)}$ , then derives a one-dimensional effective forward PDE.
- Matching that PDE with the SABR forward PDE gives effective parameters  $\alpha_{\text{eff}}, \rho_{\text{eff}}, \nu_{\text{eff}}$ .

## 5. Key intuitions

- **Cross FX dynamics:** common moves versus the base currency partly cancel in the ratio.
- **Effective volatility:** the cross does not need a new independent factor; its variance comes from combining the two liquid factors and their correlation.
- **Why an effective SABR emerges:** for European options, the key object is the marginal law of  $\tilde{X}_T$ , not the full joint law with all volatility factors.
- **Purpose of the asymptotic reduction:** integrate out hidden volatility variables and replace the original high-dimensional model by a one-dimensional forward equation with SABR-type coefficients.
- **Central closure idea in Appendix A:** the conservation law for  $Q^{(0)}$  is exact, but it depends on  $Q^{(2)}$ ; the asymptotics provide a constitutive relation  $Q^{(2)} \approx \text{coefficient} \times Q^{(0)}$ .

## 6. What I am still working on

- Some detailed asymptotic manipulations in Appendix A.
- The exact derivation of the correction terms in the general multifactor case.
- The maturity-dependent correction entering  $\alpha_{\text{eff}}$ .
- The full general-correlation formulas of Appendix B.
- The calibration and implementation implications in practice.